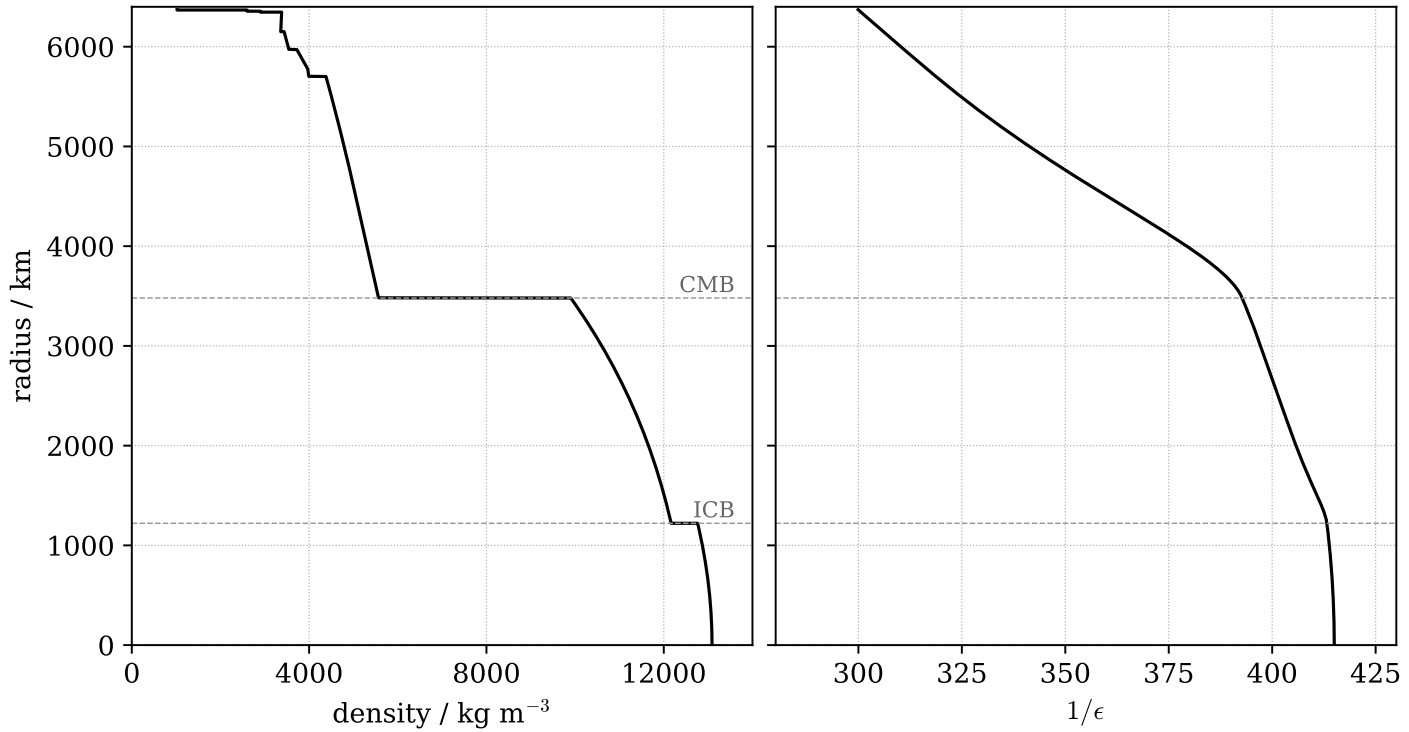
Lecture 23

Free oscillations

From last time…



Dziewonski & Anderson (1981)

Operator formalism

Linearised equations with self-gravitation and rotation:

$$\rho\left(\frac{\partial^2 u_i}{\partial t^2}+2\epsilon_{ijk}\Omega_j\frac{\partial u_k}{\partial t}+\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m\right)-\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right)-\rho\gamma_i^1=-\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

$$\gamma_i^1=-G\int_M\rho'\left(\frac{\delta_{ij}}{[(x_k-x_k')(x_k-x_k')]^{3/2}}-\frac{3(x_i-x_i')(x_j-x_j')}{[(x_k-x_k')(x_k-x_k')]^{5/2}}\right)(u_j-u_j')\,\mathrm{d}^3\mathbf{x}',$$

$$A_{ijkl}\frac{\partial u_k}{\partial x_l}\hat{n}_j=\overline{S}_{ij}\hat{n}_j.$$

Simplifying notations will help a lot!

For an arbitrary **test function**, we have:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) - \rho\gamma_i^1 \right] w_i^* \, \mathrm{d}^3\mathbf{x} = 0.$$

Integrating by parts and using the boundary conditions:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl}\frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] \mathrm{d}^3\mathbf{x} = 0.$$

If the latter identity holds for all test functions, then we can recover both the equations of motion and the boundary conditions. This is the **weak formulation** of the problem.

Weak and strong formulations are equivalent up to technicalities.

We define the **kinetic energy form**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i \, \mathrm{d}^3\mathbf{x}.$$

The acceleration term in the weak form can then be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle.$$

This **sesquilinear form** is associated with the linear operator:

$$(P\mathbf{u})_i = \rho u_i.$$

We define the **Coriolis form**:

$$\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, \mathrm{d}^3\mathbf{x},$$

associated with the linear operator:

$$(W\mathbf{u})_i = 2\rho \epsilon_{ijk} \Omega_j u_k.$$

We define the **potential energy form**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) \, \mathrm{d}^3 \mathbf{x},$$

along with the linear form linked to the stress glut:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \overline{S}_{ij} \frac{\partial w_i^*}{\partial x_j} \, \mathrm{d}^3 \mathbf{x}.$$

These two terms incorporate **both** the equations of motion and the boundary conditions.

The weak form of the equations of motion can now be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

which is to hold for all test functions, $\mathbf{w}$.

We can formally think of the problem as

$$P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f},$$

but the boundary conditions are then left implicit.

The kinetic energy form is **self-adjoint**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \langle \mathbf{u} | P | \mathbf{w} \rangle^*.$$

It is also positive for non-zero $\mathbf{u}$, and so defines an **inner product**.

The Coriolis form is **anti-self-adjoint**:

$$\begin{aligned} \langle \mathbf{w} | W | \mathbf{u} \rangle &= \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, \mathrm{d}^3\mathbf{x} \\ &= -\left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, \mathrm{d}^3\mathbf{x} \right)^* = -\langle \mathbf{u} | W | \mathbf{w} \rangle^*. \end{aligned}$$

The potential energy form is **self-adjoint**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{u} | H | \mathbf{w} \rangle^*.$$

Elastic term is clear due to the hyperelastic symmetry, $A_{ijkl} = A_{klij}$.

Centrifugal term follows by a simple calculation – details in the handout.

Gravitational term takes more work but proceeds in the same manner – details are not in the handout and are not examinable.

Non-rotating eigenvalue problem

For long-period seismology, rotation is a relatively small term that is often handled through perturbation theory.

Weak form in a non-rotating earth model:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle.$$

Free oscillations are time-harmonic solutions with no force term:

$$u_i(\mathbf{x}, t) = s_i(\mathbf{x}) \mathrm{e}^{\mathrm{i}\omega t},$$

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

This is an eigenvalue problem, with the squared eigenfrequency as the eigenvalue.

You need to know how to show that:

1) Squared eigenfrequencies are real
2) Eigenfunctions in distinct eigenspaces are orthogonal:

$$\langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle = 0.$$

Further properties to know but without proof:

1) Eigenspaces are all finite-dimensional
2) Countably many squared eigenfrequencies, bounded from below and accumulating only at infinity
3) Eigenfunctions form a **complete basis** for vector fields

We use k to label the eigenspaces, and m the eigenfunctions in each eigenspace.

Within each eigenspace, we can choose an orthonormal basis, and hence:

$$\langle km|P|k'm'\rangle = \delta_{kk'}\delta_{mm'}.$$

Completeness of the eigenfunctions means that we can write:

$$\mathbf{u} = \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle.$$

This is the **eigenfunction expansion** of the vector field.

Equations of motion in the frequency domain:

$$-\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle.$$

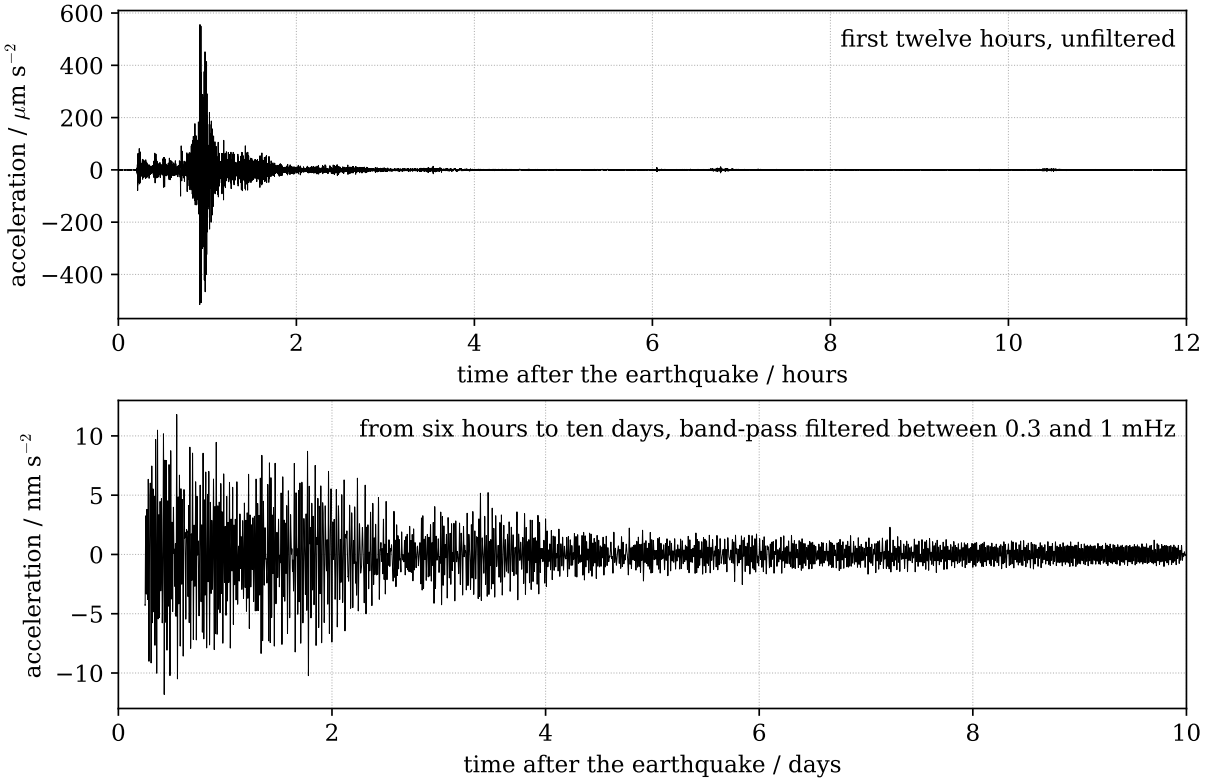
Expanding the solution in the eigenfunction basis:

$$\tilde{\mathbf{u}} = \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle | k'm' \rangle.$$

Frequency-domain solution readily derived:

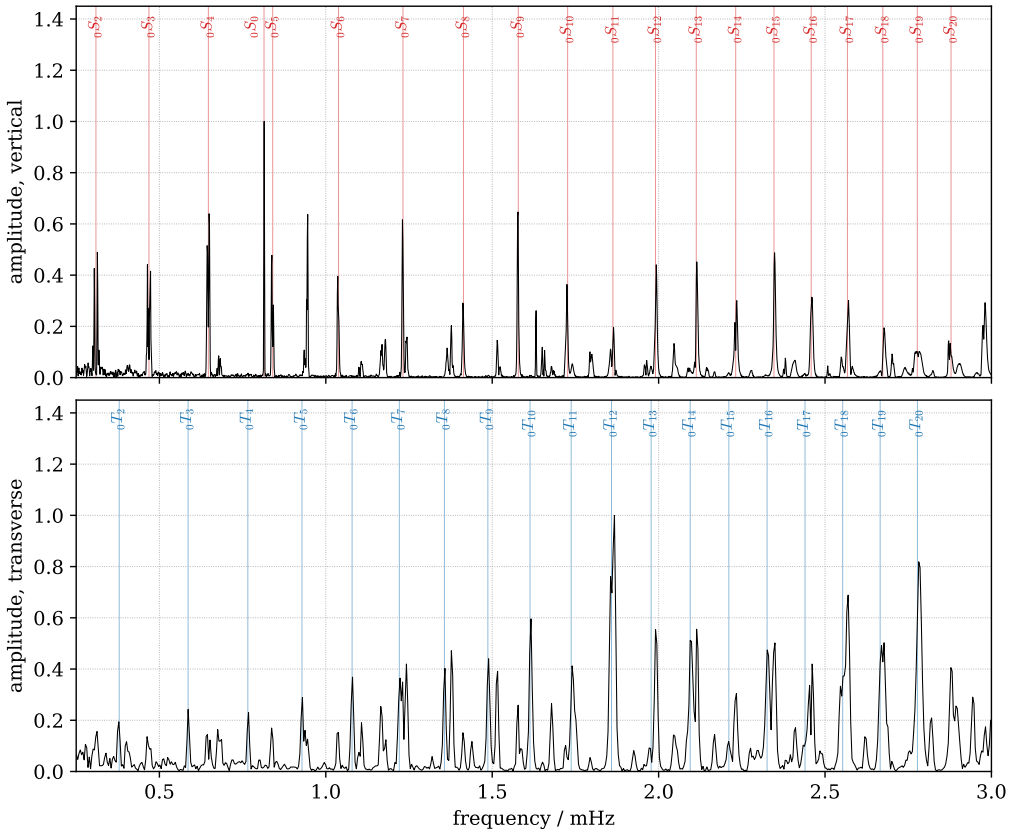
$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} | km \rangle.$$

Time-domain solution follows by contour integration – not examinable, but you will use it in the second problem set.



Ten-day record following a large earthquake (Tohoku 2011, recorded in the Black Forest, Germany).

The effects of attenuation are clearly seen within real data.



Resonance peaks (or poles) associated with each eigenfrequency; vertical lines mark those of PREM.

The eigenfrequencies depend only on Earth structure, avoiding the source–structure trade-off.

A real record has a finite length, T:

$$\mathbf{u}_{\mathrm{obs}}(t) = h(t)\mathbf{u}(t).$$

Its spectrum is a smoothed version of the true one:

$$\tilde{\mathbf{u}}_{\mathrm{obs}}(\omega) \propto \int_{-\infty}^{\infty} \tilde{\mathbf{u}}(\omega')\tilde{h}(\omega - \omega')\,\mathrm{d}\omega'.$$

The resonance peaks are broadened over a width of order T^{-1}.

For good resolution in the frequency domain, we need **long time series** – and hence the largest earthquakes.

Gravitational stability

The non-rotating eigenvalue problem:

$$-\omega_k^2 \langle \mathbf{w} | P | km \rangle + \langle \mathbf{w} | H | km \rangle = 0.$$

Taking the test function to equal the eigenfunction:

$$\omega_k^2 = \langle km | H | km \rangle.$$

Sign of the squared eigenfrequency determined by the potential energy form.

Positive values imply **stable** oscillations about equilibrium.

Negative values imply **unstable**, exponentially growing perturbations.

Zero values correspond to **linear growth in time** (double pole at the origin).

In the non-gravitating case, the potential energy form simplifies such that:

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) \, \mathrm{d}^3 \mathbf{x}.$$

This vanishes for linearised rigid body motions, with B_{ij} anti-symmetric:

$$u_i(\mathbf{x}) = a_i + B_{ij} x_j.$$

Six-dimensional eigenspace at zero frequency, but not excited by earthquakes:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \overline{S}_{ij} \frac{\partial w_i^*}{\partial x_j} \, \mathrm{d}^3 \mathbf{x}.$$

In the non-gravitating case, the potential energy form simplifies such that:

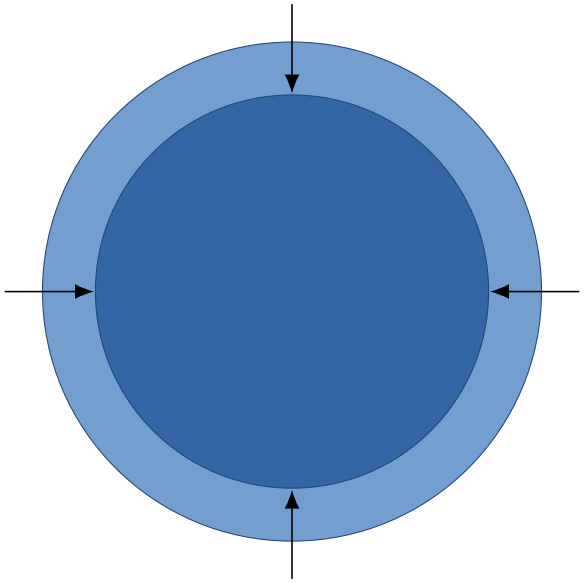
$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) \, \mathrm{d}^3\mathbf{x}.$$

Non-negative values so long as we have **uniform pointwise stability**:

$$A_{ijkl}(\mathbf{x}) e^*_{ij} e_{kl} \geq c_0 \, e^*_{ij} e_{ij}, \quad c_0 > 0.$$

Implies elastic wave speeds are all positive.

In the isotropic case, it also implies that the P-wave speed is greater than the S-wave speed.



Quantitative treatment possible, but difficult; a simple case is in the second problem set.

Stability depends on the interplay between gravitational and elastic forces.

Elastic forces always resist further deformation.

Self-gravitation is **inherently destabilising**.

A spherically symmetric compression would lower the (always negative) gravitational binding energy, while raising the elastic potential energy.

For realistic earth models such as PREM, calculations find no negative squared eigenfrequencies.

What you need to know and be able to do

(i) The weak formulation of the equations of motion using bra-ket-like notation, and the properties of the different sesquilinear forms.

(ii) How to derive the reality of the squared eigenfrequencies and orthogonality of eigenfunctions in distinct eigenspaces. State without proof the remaining important properties of the eigenvalue problem.

(iii) How to obtain a solution of the forced equations of motion in a non-rotating earth model as an eigenfunction expansion in the frequency domain. You do not need to know how to recover the time-domain solution, but may be asked to verify that such a solution works.

(iv) That eigenfrequencies are associated with resonance peaks in the amplitude spectra obtained from seismograms, and why it is necessary to look at data from large earthquakes to obtain adequate resolution.

(v) What it means for a non-rotating earth model to be stable in terms of the eigenfrequencies. You should understand the quantitative argument for the non-gravitating case, and the qualitative one when gravity is included.